XD

mercoledì 4 febbraio 2026 11:25

$$|\{S \subseteq [9] : \overset{|A|}{2 \in S} \text{ o } \overset{|B|}{\{4,5\} \cap S = \emptyset}\}|$$

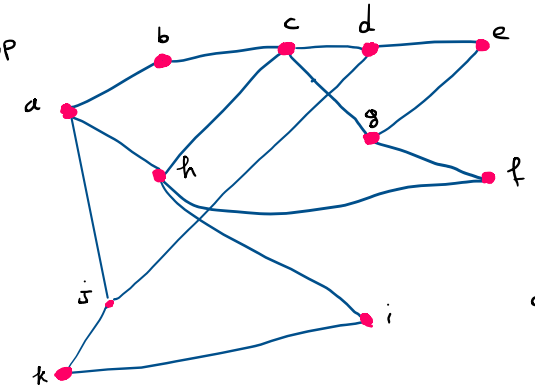
$$|A \cup B| = |A| + |B| - |A \cap B| = 2^8 + 2^7 - 2^6 = 256 + 128 - 64 = 320$$

$|A| = 2^8$

$|B| = 2^7$

$|A \cap B| = 2^6$

OP



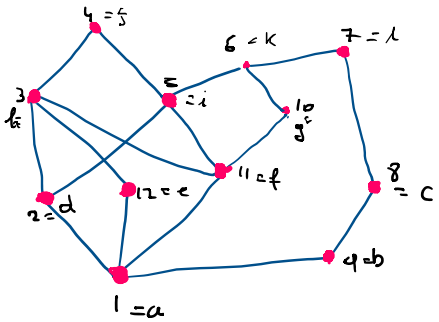
il tempo è 5

n° processori 3

dp:	0	1	2	3	4
	a	b	c	d	e
	k	h	f	g	
	j	i			

op. di tempo minimo è

$(\{a,k\},\{b,h,j\},\{c,f,i\},\{d,g\},\{e\})$



ordinamento topologico

$(1, 9, 8, 2, 12, 11, 10, 3, 5, 4, 6, 7)$

$$\sum_{k=1}^{n} (k - \cos(k)) =$$

$\sin k > -1 \quad \forall k \in \mathbb{N}$

$$(k - \cos(k))' = 1 - (-\sin(k)) = 1 + \sin k \geq 0$$

$$f(1) + \int_1^n f(x)dx \geq S_n \geq f(n) + \int_1^n f(x)dx$$

$$\int_1^n (k - \cos(k)) = \int_1^n x\,dx - \int_1^n \cos(x)\,dx = \left[\frac{x^2}{2} - \sin(x)\right]^n$$

$$\int_1 (k-\cos(k)) = \int_1 x\,dx - \int_1 \cos(k)\,dx = \left[\frac{x^2}{2} - \sin(x)\right]$$

$$\frac{m^2}{2} - \sin(m) - \left(\frac{1}{2} - \sin(1)\right) =$$

$$1 + \cos(1) + \frac{m^2}{2} - \sin(m) - \frac{1}{2} + \sin(1) \geq \frac{\sum_{k=1}^{m}(k-\cos(k))}{\frac{m^2}{2}} \geq 1 - \cos m + \frac{m^2}{2} - \sin(m) - \frac{1}{2} + \sin(1)$$

$$\frac{1}{\frac{m^2}{2}} \to 0 \qquad \frac{\cos(1)}{\frac{m^2}{2}} \to 0 \qquad \frac{\frac{m^2}{2}}{\frac{m^2}{2}} = 1 \qquad \frac{-\sin(m)}{\frac{m^2}{2}} \to 0 \qquad \frac{-\frac{1}{2}}{\frac{m^2}{2}} \to 0 \qquad \frac{\sin(1)}{\frac{m^2}{2}} \to 0$$

$$\frac{\cos m}{\frac{m^2}{2}} \to 0 \qquad \frac{\cos m}{\frac{m^2}{2}} \to 0 \qquad \frac{\frac{m^2}{2}}{\frac{m^2}{2}} = 1 \qquad \frac{-\sin(m)}{\frac{m^2}{2}} \to 0 \qquad \frac{-\frac{1}{2}}{\frac{m^2}{2}} \to 0 \qquad \frac{\sin(1)}{\frac{m^2}{2}} \to 0$$

Per confronto sono che

$$1 \geq \frac{\sum_{k=1}^{m}(k-\cos(k))}{\frac{m^2}{2}} \geq 1$$

$$\lim_{m\to\infty} \frac{\sum_{k=1}^{m}(k-(\cos(k)))}{\frac{m^2}{2}} = 1$$

Perciò

$$\sum_{k=1}^{m}(k-(\cos(k))) \sim \frac{m^2}{2}$$

$$\sum_{k=1}^{m} 2k\ln(k) = S_m$$

$$(2x\ln x)' = 2\ln x + 2x\frac{1}{x} = 2(\ln(x)+1)$$

$\ln(x) > -1 \quad \ln(x) > \ln e^{-1} \to x > \frac{1}{e}$

è crescente

$$f(1) + \int_1^m 2x\ln(x)\,dx \geq S_m \geq f(m) + \int_1^m 2x\ln(x)\,dx$$

$$\int_1^m 2x\ln x\,dx = 2\left(\frac{x^2}{2}\ln(x) - \int \frac{x^2}{2}\frac{1}{x}\,dx\right) = 2\left(\frac{x^2}{2}\ln(x) - \frac{1}{2}\int x\,dx\right)$$

$$2\left(\frac{x^2}{2}\ln(x) - \frac{1}{2}\frac{x^2}{2}\right) + c \Rightarrow \left[x^2\ln(x) - \frac{x^2}{2}\right]_1^m = m^2\ln(x) - \frac{m^2}{2} - \left(0 - \frac{1}{2}\right) =$$

$$= m^2\ln(x) - \frac{m^2}{2} + \frac{1}{2} = m^2\left(\ln(x) - \frac{1}{2}\right) + \frac{1}{2}$$

$$0 + m^2\left(\ln(m) - \frac{1}{2}\right) + \frac{1}{2} \geq \frac{S_m}{a_m} \geq \frac{2m\ln(m)}{a_m} + \frac{m^2\left(\ln(m) - \frac{1}{2}\right)}{a_m} + \frac{\frac{1}{2}}{a_m}$$

$$a_m = m^2\left(\ln(m) - \frac{1}{2}\right)$$

$$1 \qquad \to 0 \qquad \to 0 \qquad 1 \qquad \to 0$$

$$\frac{2m\ln(m)}{m^2(\ln(m) - \frac{1}{2})} = 0$$

dunque: $1 \geq \frac{\sum_{k=1}^{m} 2k\ln(k)}{m^2(\ln(m) - \frac{1}{2})} \geq 1$ Per confronto

$$\frac{2n\ln(n)}{n^2(\ln(n)+\frac{1}{2})} = 0 \qquad \Rightarrow \qquad n^2(\ln(n)+\tfrac{1}{2})$$

$$\lim_{n\to+\infty} \frac{\sum_{k=1}^{n} 2k\ln(k)}{n^2(\ln(n)-\frac{1}{2})} = 1 \quad \text{dunque}$$

$$\sum_{k=1}^{n} 2k\ln(k) \sim n^2\left(\ln(n)-\tfrac{1}{2}\right)$$

$$\sum_{k=1}^{n} 3k^2-3k+1 =: S_n$$

$$(3k^2-3k+1)' = 3(2)k-3) = 6k-3 = 3(2k-1)$$

$$\text{dunque} \quad 2k-1>0 \to k>\tfrac{1}{2} \quad \text{ok}$$

$$f(n) + \int_1^n f(x)\,dx \geq S_n \geq +f(n) + \int_1^n f(x)\,dx$$

$$\int_1^n 3x^2-3x+1\,dx = 3\int x^2\,dx - 3\int x\,dx + \int 1\,dx = \left[\frac{\cancel{3}x^3}{\cancel{3}} - \frac{3x^2}{2} + x\right]_1^n$$

$$n^3 - \frac{3n^2}{2} + n - \left(1 - \frac{3}{2} + 1\right) = n^3 - \frac{3n^2}{2} + n - \frac{1}{2}$$

$$2-\frac{3}{2} = \frac{4-3}{2} = \frac{1}{2}$$

$$\cancel{3} - \cancel{3} + \underbrace{\cancel{1}}_{\to 0} + \underbrace{\left(n^3 - \frac{3n^2}{2} + n - \frac{1}{2}\right)}_{\sim 1} \geq \underbrace{S_n}_{\sim 1} \geq \underbrace{\left(\cancel{3n^2} - \cancel{3n} + 1\right)}_{\to 0} + \underbrace{\left(n^3 - \frac{3n^2}{2} + n - \frac{1}{2}\right)}_{\sim 1}$$

$$\text{dunque}$$

$$1 \geq \frac{S_n}{\sim 1} \geq 1 \implies \lim_{n\to+\infty} \frac{\sum_{k=1}^{n} 3k^2-3k+1}{n^3 - \frac{2n^2}{2} + n - \frac{1}{2}} = 1$$

$$\sum_{k=1}^{n} 3k^2-3k+1 \sim n^3 - \frac{2n^2}{2} + n - \frac{1}{2}$$

OSS volendo anche

$$\sum_{k=1}^{n} 3k^2-3k+1 \sim n^3$$

$$\sum_{k=1}^{n} k^4 =$$

Sappiamo dalla teoria che $\sum_{k=1}^{n} k^4 = An^5 + Bn^4 + Cn^3 + Dn^2 + En + F$ (*)

$n=0 \Longrightarrow F=0$

$n=1 \Longrightarrow A+B+C+D+E+F=1$

$n=2 \Longrightarrow A32+B16+C8+D4+E2+F=17$

$n=3 \Longrightarrow A243+B81+C27+D9+E3+F=98$

$n=4 \Longrightarrow A1024+B256+C64+D16+E4+F=354$

$n=5 \Longrightarrow A3125+B625+C125+D25+E5+F=979$

Risolvendo il sistema e trovato i coeff. e sostituendo su (*) si ha la soluz. richiesta

Coefficiente	Valore
A	1/5
B	1/2
C	1/3
D	0
E	-1/30

ovvero

$$\sum_{k=1}^{n} k^4 = \frac{n^5}{5} + \frac{n^4}{2} + \frac{n^3}{3} - \frac{n}{30}$$

Stimare $n!$

Sappiamo che

$$n! = \prod_{k=1}^{n} k \Longrightarrow \ln\left(\prod_{k=1}^{n} k\right) \Longrightarrow \sum_{k=1}^{n} \ln(k) \Rightarrow \exp\left(\sum_{k=1}^{n} \ln(k)\right)$$

$$\sum_{k=1}^{n} \ln(k) =: S_n \qquad (\ln(x))' = \frac{1}{x} > 0$$

$$\ln 1 + n\ln(n) - n + 1 \geq S_n \geq \ln n + n\ln n - n + 1$$

$$\Downarrow$$

$$\exp(\ln 1 + n\ln(n) - n + 1) \geq n! \geq \exp(\ln n + n\ln n - n + 1)$$

$$\exp(\ln n^n - n + 1) \geq n! \geq \exp(\ln(n) + \ln n^n - n + 2)$$

$$n^n \frac{1}{e^n} \cdot e \geq n! \geq n \cdot n^n \frac{1}{e^n} \cdot e \quad \leftarrow \text{stima}$$

Sia $|\{S=[50] : n|3 \text{ o } n|5\}|$

A: B:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$|A \cup B| = |A| + |B| - |A \cap B|$

$|A| = \frac{50}{3} = 16+1$ $\quad |A \cap B| = \{15, 30, 45, 0\}$

$|B| = \frac{50}{5} = 10+1$

$|A \cup B| = 17 + 11 - 4 = 24$